



LOB 1036 - Geometria Analítica

Lista de exercícios 3 - Parte 2
2º semestre de 2025

1.

a) $x^2 = 8y$

b) $y^2 = 8x$

c) $x^2 = -4y$

d) $x^2 = -\frac{4}{3}y$

e) $x^2 + 4x + 8y - 20 = 0$

f) $y^2 - 2y - 32x - 129 = 0$

g) $y^2 = \frac{25}{4}x$

h) $x^2 + 8x + 8y - 8 = 0$

i) $x - 12x - 12y + 48 = 0$

j) $(y+1)^2 = 5\left(x - \frac{7}{4}\right)$

k) $(y-3)^2 - 8(x-1)$

l) $y = -\frac{x^2}{3} + \frac{4x}{3}$

2.

a) $V(0,0), F(0,-3), d: y=3, e: x=0$

b) $V(0,0), F(-25,0), d: x=25, e: y=0$

c) $V(-2,-1), F(-2,-3), d: y=1, e: x=-2$

d) $V(1,-2), F(1,3), d: y=-7, e: x=1$

e) $V(3,-1), F(-1,-1), d: x=7, e: y=-2$

3.

a) $C(0,0), A(\pm 10,0), F(\pm 8,0), e = \frac{4}{5}$

b) $C(0,0), A(0,\pm 10), F(0,\pm 8), e = \frac{4}{5}$

c) $C(0,0), A(0, \pm 3), F(0, \pm 2), e = \frac{2}{3}$

d) $C(0,0), A(0, \pm 1), F\left(0, \pm \frac{\sqrt{3}}{2}\right), e = \frac{\sqrt{3}}{2}$

e) $C(0,0), A\left(\pm \frac{1}{2}, 0\right), F\left(\pm \frac{\sqrt{21}}{10}\right), e = \frac{\sqrt{21}}{5}$

f) $C(0,0), A\left(\pm \frac{5}{3}, 0\right), F\left(\pm \frac{4}{3}, 0\right), e = \frac{4}{5}$

4.

a) $\frac{x^2}{25} + \frac{y^2}{9} = 1$

b) $\frac{y^2}{9} + \frac{x^2}{4} = 1$

c) $\frac{x^2}{36} + \frac{y^2}{9} = 1$

d) $\frac{x^2}{9} + \frac{y^2}{5} = 1$

e) $7x^2 + 16y^2 - 28x - 128y + 172 = 0$

f) $5x^2 + 9y^2 + 30x = 0$

g) $9x^2 + 16y^2 + 54x - 128y + 193 = 0$

h) $x^2 + 9y^2 + 8x - 36y + 43 = 0$

5.

a) $C(2, -3), A_1(-2, -3), A_2(6, -2), F_1(2 - \sqrt{7}, -3), F_2(2 + \sqrt{7}, -3), e = \frac{\sqrt{7}}{4}$

b) $C(-1, -2), A_1(-1, -7), A_2(-1, 3), F_1(-1, -5), F_2(-1, 1), e = \frac{3}{5}$

c) $C(3, -1), A_1(0, -1), A_2(6, -1), F_1(3 - \sqrt{5}, -1), F_2(3 + \sqrt{5}, -1), e = \frac{\sqrt{5}}{3}$

d) $C(-2, 2), A_1(-2, -2), A_2(-2, 6), F_1(-2, 2 - \sqrt{15}), F_2(-2, 2 + \sqrt{15}), e = \frac{\sqrt{15}}{4}$

e) $C(3, -4), A_1(3, -8), A_2(3, 0), F_1(3, -4 - \sqrt{7}), F_2(3, -4 + \sqrt{7}), e = \frac{\sqrt{7}}{4}$

f) $C(1, 2), A_1(-2, 2), A_2(4, 2), F_1(1 - \sqrt{5}, 2), F_2(1 + \sqrt{5}, 2), e = \frac{\sqrt{5}}{3}$

6.

a) $A(\pm 10, 0), F(\pm 2\sqrt{41}, 0), e = \frac{\sqrt{41}}{5}$

b) $A(0, \pm 10), F(0, \pm 2\sqrt{41}), e = \frac{\sqrt{41}}{5}$

c) $A(0, \pm 2), F(0, \pm 3), e = \frac{3}{2}$

d) $A(\pm 2\sqrt{2}, 0), F(\pm 2\sqrt{3}, 0), e = \frac{\sqrt{6}}{2}$

$$\text{e) } A(0, \pm 1), F\left(0, \pm \frac{\sqrt{5}}{2}\right), e = \frac{\sqrt{5}}{2}$$

$$\text{f) } A\left(0, \pm \frac{\sqrt{2}}{2}\right), F\left(0, \pm \frac{\sqrt{3}}{2}\right), e = \frac{\sqrt{6}}{2}$$

7.

$$\text{a) } \frac{x^2}{9} - \frac{y^2}{16} = 1$$

$$\text{b) } x^2 - 12y^2 - 16 = 0$$

$$\text{c) } \frac{x^2}{9} - \frac{y^2}{36} = 1$$

$$\text{d) } 3y^2 - x^2 - 12y + 10x - 40 = 0$$

$$\text{e) } x^2 - y^2 - 10x + 2y + 16 = 0$$

$$\text{f) } 8y^2 - 5x^2 + 48y + 20x + 25 = 0$$

$$\text{g) } 24x^2 - 5y^2 + 96x + 10y = 0$$

8.

$$\text{a) } C(1, -2), A_1(-1, -2), A_2(3, -2), F_1(1 - \sqrt{13}, -2), F_2(1 + \sqrt{13}, -2), e = \frac{\sqrt{13}}{2}$$

$$\text{b) } C(-3, 3), A_1(-5, 3), A_2(-1, 3), F_1(-3 - \sqrt{5}, 3), F_2(-3 + \sqrt{5}, 3)$$

$$\text{c) } C(3, 1), A_1(3, -2), A_2(3, 4), F_1(3, 1 - \sqrt{13}), F_2(3, 1 + \sqrt{13}), e = \frac{\sqrt{13}}{3}$$

$$\text{d) } C(4, 2), A_1(1, 2), A_2(7, 2), F_1(4 - 3\sqrt{5}, 2), F_2(4 + 3\sqrt{5}, 2), e = \sqrt{5}$$

$$\text{e) } C(-2, 3), A_1(-2, -3), A_2(-2, 9), F_1(-2, 3 - 2\sqrt{10}), F_2(-2, 3 + 2\sqrt{10}), e = \frac{\sqrt{10}}{3}$$

$$\text{f) } C(2, -1), A_1(2, -5), A_2(2, 3), F_1(2, -6), F_2(2, 4), e = \frac{5}{4}$$